Research activity

Three lines facing a plane — 30 min.

In an orthonormal frame of space, consider the plane $\mathcal{P}$ with equation $x + y + z = 3$. In each question, the parameter t ranges over $\mathbb{R}$.

1. The line d_1 has the representation $(x, y, z) = (t, t, t)$. Determine $d_1 \cap \mathcal{P}$.
2. The line d_2 has the representation $(x, y, z) = (3 + t, -t, 0)$. Determine $d_2 \cap \mathcal{P}$.
3. The line d_3 has the representation $(x, y, z) = (t, -t, 0)$. Determine $d_3 \cap \mathcal{P}$.
4. For each line, state whether the equation obtained after substitution into that of the plane has a unique solution, no solution, or every real number as a solution. Link these cases to the geometric positions found.